

Lecture 5

The Labor Market

Economics B — Macroeconomics I

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Learning Objectives and Literature of the 5th Lecture

- ▶ Learning objectives:
 - ▶ Transition from the short- to the medium-run: scarce resources
 - ▶ Labor supply and demand
 - ▶ Types and causes of unemployment
 - ▶ The labor market in an economic model
 - ▶ State interventions in the labor market
- ▶ Literature: Textbook by Blanchard, chapter 7

Differences between short- and medium-run

- ▶ In our analysis of the economy in the short-run, we neglected prices and resource scarcity.
- ▶ The production level was then determined by demand factors such as consumer spending or monetary and fiscal policy.
 - ▶ Implicitly we ignored the production side in the IS-LM model.
“Demand determines production.”
 - ▶ → Reversal of Say’s law
- ▶ Theoretically, in the IS-LM model, any high level of production can be achieved through expansionary fiscal and monetary policy.
Obviously, the model is incomplete and suitable only for deep recessions.

Limits of the short-run analysis

- ▶ Expansionary fiscal and monetary policy reach their limits in the medium run:
 - ▶ The labor market lacks suitable personnel for additional government spending ($G \uparrow$, such as infrastructure programs).
 - ▶ In the short-run, employees accept extra hours, in the medium-run overtime is reduced or paid out (wages, prices $\uparrow$)
 - ▶ Massive tax cuts increase the disposable income (Y_D) and thus the demand for goods, but the supply increases only to a limited extent (see previous points). As a result, firms raise prices $\rightarrow$ inflation.
 - ▶ If the state takes on debt for expansionary fiscal policy, taxes must be raised in the medium run (or debt 'inflated away').
 - ▶ Likewise, increasing prices (i.e., inflation) will occur in the medium-run, if the central bank significantly expands the money supply.
- ▶ Application: German special fund (Sondervermögen)

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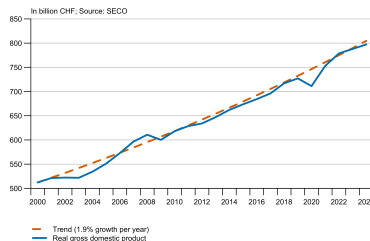
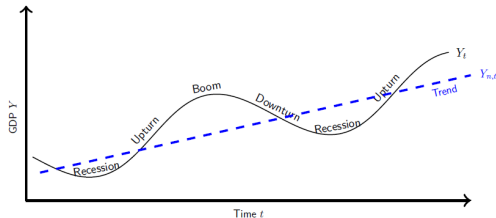
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Resource limits in the medium-run

- ▶ In the medium-run, we must extend our analysis to include the labor market and inflation.
- ▶ Scarce resources:
 - ▶ In the short-run, demand determines the GDP. Factors of production (labor, capital) may be underutilized / excessively burdened.
 - ▶ In the medium-run, the economy returns to an equilibrium in which production equals the output capacity.
- ▶ We focus on the shortage of labor (as a production factor).

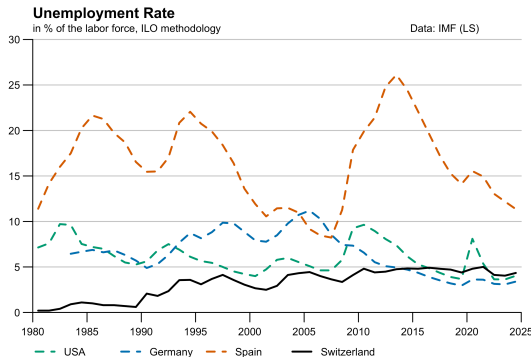
Supply and demand perspective

- Change from the demand to the supply perspective:
We now examine the level of the trend line at a given point in time.
In lectures 10 & 11, we will discuss the slope of the trend line.



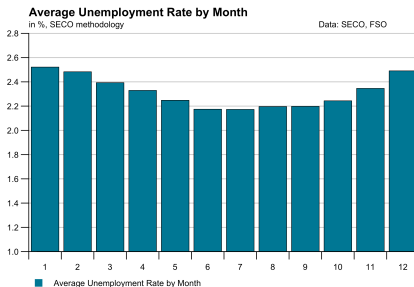
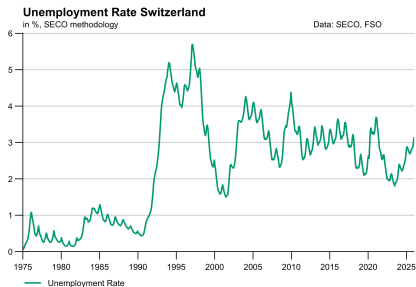
Stylized facts about the labor market

- ▶ Unemployment is considered one of the biggest economic problems.
- ▶ Why do unemployment rates vary so much across countries and time?
- ▶ Why does the unemployment rate never drop to zero?



Stylized facts about the labor market — Switzerland

► Why does the Swiss unemployment rate vary over time?



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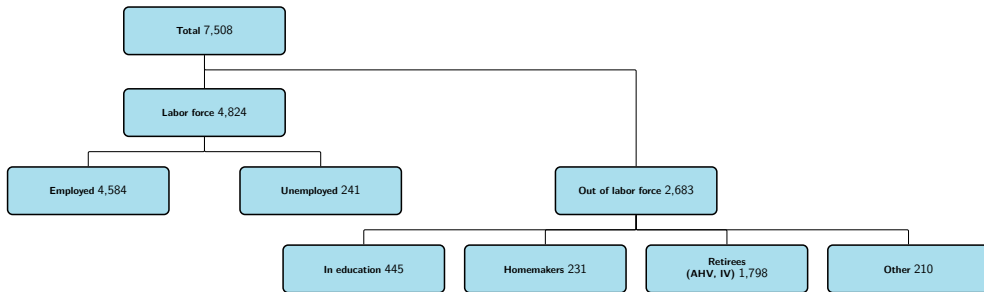
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The labor market

- Who participates in the labor market (permanent resident population aged 15+)?



in 1,000 persons, Source: FSO Labor market status by sociodemographic characteristics, 2024

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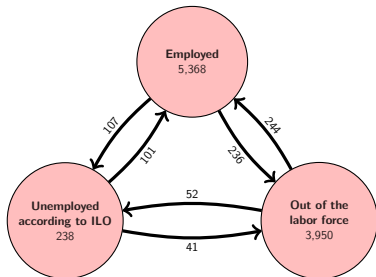
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Participants in the labor market

- ▶ Employment rate = employed people / reference population
- ▶ Participation rate = civilian labor force / reference population
 - ▶ Reference population: total population, population age 16 or older, often: Civilian Noninstitutional Population (age 16+, not in military, institutions)
- ▶ Unemployment rate = $1 - \text{employment rate}$
- ▶ There are different ways to measure the unemployment rate (cf. textbook p. 28). In Switzerland, the unemployment rate according to ILO is about twice as high as the unemployment rate according to Seco.

Labor market dynamics

- ▶ Unemployment exists, because not every unemployed person can immediately get a job. Nevertheless, there is a great dynamic.
- ▶ Many people change their status in the labor market over the course of a year (figures in 1,000)



Stocks as of Dec 31, 2024, flows from Jan 1, 2024 to Dec 31, 2024 (in 1,000). Source: Federal Statistical Office, 2024

Inflow, outflow, and the unemployment rate

- ▶ A simple model of labor market dynamics:
 - ▶ Assuming a probability p of losing the job, and a chance of finding a job s :



Inflow, outflow, and the unemployment rate (continued)

- ▶ The number of unemployed is constant when the following condition holds:

Entry into unemployment = exit from unemployment

$$\Longleftrightarrow p * N = s * U$$

- ▶ Re-write with *labor force* $L = N + U$:

$$p * (L - U) = s * U$$

$$\Longleftrightarrow p * (L/L - U/L) = s * U/L$$

with $u = U/L$ it follows:

$$\Longleftrightarrow p * (1 - u) = s * u$$

$$\Longleftrightarrow u^* = p / (p + s)$$

- ▶ Numerical example: $p = 1.5\%$, $s = 20\%$, $\rightarrow u = 0.015 / 0.215 = 7.0\%$

Inflow, outflow, and the unemployment rate – Empirics

- Average monthly values of Elsby et al. (*REStat*, 2013), for Switzerland (ILO) figures for 2013:

	Rate of unemployment (u)	Outflow Rate (s)	Inflow Rate (p)
Australia	7.1	22.8	1.7
Germany	8.3	6.0	0.5
France	8.1	7.7	0.7
Japan	3.3	18.9	0.6
Switzerland	4.4	33.0	1.5
US	6.1	56.5	3.6
UK	7.7	13.9	1.0

- Differences are partly explained by labor protection laws like notice periods or severance payments:
 - Reducing the likelihood of losing your job (p), but also the probability of finding a job (s).
- Search-and-matching models start here and explain p and s .

Types of unemployment

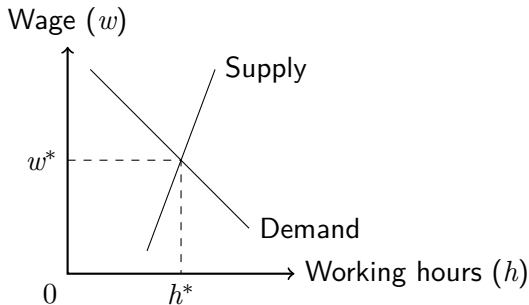
- ▶ Before we develop an economic model for the labor market, first a discussion of the types & causes of unemployment.
- ▶ Types of unemployment:
 - ▶ Frictional, seasonal, cyclical, structural
- ▶ Remember:
 - ▶ Bear in mind not only the unemployment rate but also the employment rate.
 - ▶ Underemployment: employed persons with involuntarily short working hours who are available for a higher workload (5% in Switzerland according to the Federal Statistical Office, 2024)
- ▶ Particularly problematic are long-run and youth unemployment.
 - ▶ → Loss or lack of human capital

Causes of unemployment

- ▶ What does the unemployment rate in a country depend on?
- ▶ Deviations from the 'perfect' labor market:
 1. Minimum wage (also implicit in unemployment benefit → reservation wage)
 2. Unions: Insider vs. Outsider Theory
 3. Market power of the firms
 4. Taxes and social contributions (→ wedge between income and costs)
- ▶ In the following model we focus on points 1 to 3.

Simple labor market model

- ▶ To study the labor market within the framework of an economic model, we could simply represent labor supply (by workers) and labor demand (by firms):



- ▶ What are the implicit assumptions here?
- ▶ Why do we (and the textbook) choose a different approach?

Labor supply

- ▶ The labor supply by individuals is determined by:
 - ▶ the decision to participate in the labor market (extensive labor supply)
 - ▶ the choice of the amount of labor supply (intensive labor supply)
- ▶ What factors explain extensive and intensive labor supply?
 - ▶ Wages, opportunity costs of work (leisure, education, childcare), labor market institutions, demography, culture, etc.

Labor supply: Determining wages

- ▶ If the wage is above the opportunity cost, work is offered.
Otherwise, someone is voluntarily unemployed.
But how does involuntary unemployment occur?
- ▶ Individual negotiations between employee and employer often emerge only for higher-qualified jobs. Instead, often collective bargaining between unions and employers' associations.
 - ▶ Wages are usually fixed for a period of 1–2 years.
 - ▶ Can take place at a firm-, industry-, or country-level.
 - ▶ Generally much more important in European than in Anglo-Saxon countries.

Labor supply: Determining wages — Formally

- ▶ Important factors in wage negotiations:
 - ▶ Expected price level (increase)
 - ▶ Bargaining power of the two sides
 - ▶ Reservation wage of employees
- ▶ We sum up with one equation: $W = P^e F(u, z)$
(-,+)
 - ▶ The nominal wage W rises in the expected price level P^e .
Relevant for the employees is the real wage W/P .
 - ▶ Nominal wages decrease the higher the unemployment rate (u).
 - ▶ The nominal wage depends on other factors (z)
such as unemployment insurance or minimum wage.

Labor demand

- ▶ Firms need workers for production.
- ▶ Assumption: firms have a production function $Y = A * N$.
There are no differences in qualification across workers.
 - ▶ Output Y is determined by productivity A and employment N .
 - ▶ Note: For simplicity, we assume a constant marginal product, so that the labor demand curve is horizontal.
 - ▶ Simplification (sometimes used): $A = 1$ so that $Y = N$
- ▶ Marginal cost of an additional unit of production = W/A .
- ▶ In *perfect* competition: price = marginal cost $\longleftrightarrow$ wage = W/A
The wage offered corresponds to the marginal product of labor.

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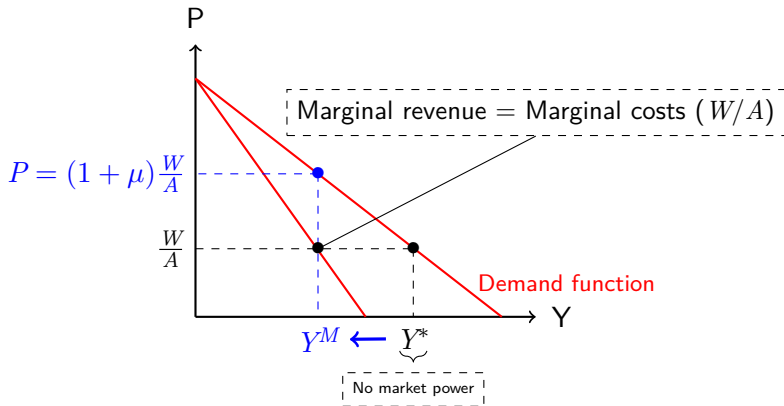
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Labor demand (continued)

- ▶ Assumption: No perfect competition in the goods market:
Price is above marginal cost: $P = (1 + \mu) W/A$, where $\mu > 0$
- ▶ Repetition from Economics A – Microeconomics: Monopoly



Labor demand (continued) — Mark-up

► Firms maximize profit $\Pi = P(Y) * Y - W/A * Y$

► In the optimum (= profit maximum) we have

$$\frac{\partial \Pi(Y)}{\partial Y} = 0 \quad \Leftrightarrow \quad \underbrace{P(Y) + \frac{\partial P(Y)}{\partial Y} * Y}_{\text{=Marginal revenue}} = \underbrace{\frac{W}{A}}_{\text{Marginal costs}}$$

$$\Leftrightarrow \quad P(Y) + P(Y) * \frac{\partial P(Y)}{\partial Y} * \frac{Y}{P(Y)} = \frac{W}{A}$$

$$\Leftrightarrow \quad P(Y) (1 + \eta_Y^P) = \frac{W}{A} \quad \Leftrightarrow \quad P(Y) \left(1 + \frac{1}{\epsilon_P^Y}\right) = \frac{W}{A}$$

► $\eta_Y^P = \frac{\partial P(Y)}{\partial Y} * \frac{Y}{P(Y)}$

► $\epsilon_P^Y = \frac{1}{\eta_Y^P} = \frac{\partial Y(P)}{\partial P} * \frac{P}{Y(P)} := \text{Price elasticity of demand}$

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Labor demand (continued) — Mark-up II

- ▶ For $\frac{W}{A} > 0$ we have $P(Y) \left(1 + \frac{1}{\epsilon_P^Y}\right) > 0 \iff \epsilon_P^Y < -1$,
- ▶ The monopolist operates on the elastic part of the demand curve
- ▶ $\mu :=$ Markup, difference between marginal costs and price (expressed in percentage)
- ▶ $P(Y) \left(1 + \frac{1}{\epsilon_P^Y}\right) = \frac{W}{A} \iff P = (1 + \mu) \frac{W}{A}$, with $\mu = -\frac{1}{1 + \epsilon_P^Y}$
 - ▶ $\epsilon_P^Y \rightarrow -1 \Rightarrow \mu \rightarrow \infty$, marginal costs are **very small** and thus the price is **very large** compared to the marginal costs.
 - ▶ $\epsilon_P^Y \rightarrow -\infty \Rightarrow \mu \rightarrow 0$, marginal costs are **very large** and thus the price is almost **identical** to marginal costs.

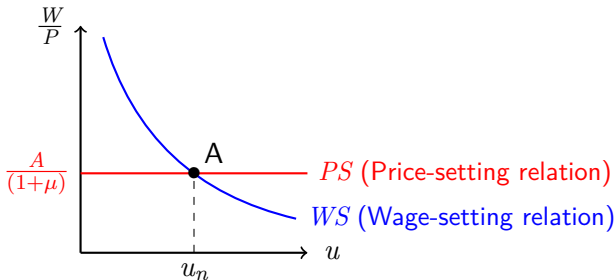
Equilibrium in the labor market

- ▶ Equilibrium in the labor market determined by supply & demand.
- ▶ Labor supply:
 - ▶ If the actual price level P corresponds to the expected price level P^e , the result is $W = P * F(u, z)$.
 - ▶ We divide both sides by P and obtain the **wage-setting relation**:
 $W/P = F(u, z)$.
- ▶ Labor demand:
 - ▶ The price level is determined by the firms' price: $P = (1 + \mu) W/A$.
 - ▶ We can re-write this and obtain the **price-setting relation**:
 $W/P = A/(1 + \mu)$.

Equilibrium in the labor market model

- ▶ We now equate wage-setting and price-setting relation.
- ▶ In the medium-run equilibrium in the labor market it holds:

$$F(u, z) = A/(1 + \mu)$$



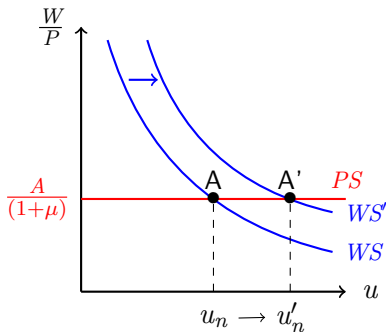
- ▶ Note: The WS -curve is drawn for a given z .

Equilibrium in the labor market model

- ▶ Why is there unemployment in equilibrium?
 - ▶ Demand side:
Firms have market power and produce 'too little.'
 - ▶ Supply side:
Wage demands by trade unions are decreasing in u , but only to a limited extent (Slope of WS -curve is not steep enough).
→ Insider-outsider problem
- ▶ What types of unemployment does the model explain?
 - ▶ The model explains structural and frictional unemployment. It considers the medium-run, in which production equals the potential output.
 - ▶ The model does not explain: seasonal or cyclical unemployment.

Comparative statics

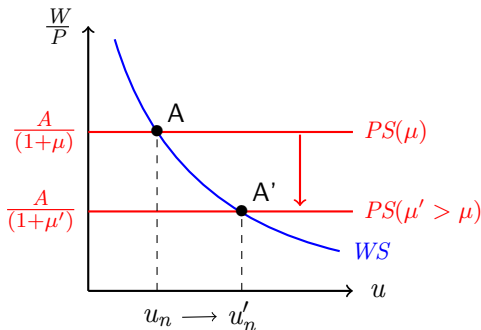
- The effects of an increase in unemployment benefits on the level of the natural rate of unemployment (u_n):



- An increase in unemployment benefits (an increase of z) leads to an increase in the natural rate of unemployment ($u_n \rightarrow u'_n$).

Comparative statics II

- The impact of a higher mark-up μ on the natural rate of unemployment:



- An increase in the markup ($\mu \rightarrow \mu'$) lowers the real wage and leads to an increase in the natural rate of unemployment.

Comparative statics III

- ▶ The government sets a binding minimum wage $W_{\min}$:
 - ▶ Supply side: $W/P = F(u, z) \rightarrow z \uparrow$.
Bargaining power of employees increases, “threat point” is higher, corresponds to higher z .
 - ▶ Demand side: $P = (1 + \mu) W/A \rightarrow$ firms increase the market price proportionally $\rightarrow W/P$ unchanged
 - ▶ PS -relation remains unchanged, WS -curve shifts up/right, in equilibrium u increases.
- ▶ Explanation:

The union demands higher wages and accepts higher unemployment. In the medium-run, the firms pass on the higher wages in the form of higher prices; they sell and produce less.
- ▶ In the medium-run, the minimum wage leads to higher prices and a higher rate of unemployment. $\rightarrow$ In the exercises we discuss *short-run* effects.

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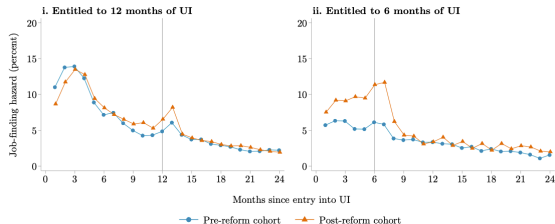
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Empirical evidence for the comparative statics

- ▶ Do empirical analyses support the statements of the model?
- ▶ An example: Hartz reforms in Germany in 2005
 - ▶ Unemployment benefits (ALG) were reduced after 12 months of unemployment to a lower level (ALG II or so-called Hartz IV).
 - ▶ Price (2025) finds a reduction of the structural unemployment rate by 0.7 percentage points as a result of the reform.



Source: Brendan M. Price (2025), "The Duration and Wage Effects of Long-Term Unemployment Benefits"

The natural rate of unemployment

- ▶ From the natural rate of unemployment follows the natural level of employment: $N_n = L * (1 - u_n)$
- ▶ This is associated with a natural level of production and output potential: $Y_n = A * N_n = A * L * (1 - u_n)$
 - ▶ This is the Y_n at a time on the trend line (previous slides)
- ▶ How 'natural' is the natural rate of unemployment u_n ?
 - ▶ The level of unemployment benefits or competition legislation can certainly not be described as 'natural.' They characterize structural framework conditions of an economy.
 - ▶ For this reason it is more appropriate to refer to the natural rate of unemployment as the structural rate of unemployment.

Political influence in the model

- ▶ How can the state reduce unemployment in the medium run and thus increase the output potential $Y_n = A * N_n$?
- ▶ According to our model:
 - ▶ Increase of A : education, capital, innovation, management
 - ▶ Lowering of μ : competition policy (*antitrust*)
 - ▶ Lowering of z : reform of unemployment insurance, childcare, tax structure

Political influence in practice

- ▶ Unemployment insurance: large variation in wage replacement rates
 - ▶ Objectives of the insurance: Short-run bridging of loss of income in case of unemployment and improvement of job-match quality
 - ▶ Problems: higher reservation wage, less search efforts
- ▶ Reduction of income tax and social security contributions
- ▶ Reduction of information asymmetries:
Internet portals, public and private job placement
- ▶ Active labor market policy: training, job creation policies
- ▶ Distribute work more evenly with early retirement and 35h a week?
→ *Lump of labor fallacy*: Wrongly assuming a fixed amount of work.
See also SECO study

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Summary and outlook

- ▶ In this chapter we have assumed that the actual price level is always equal to the expected price level.
 - ▶ In the short-run, however, the price level may differ from the price level expected at the time of wage-setting. The unemployment rate then also deviates from the natural rate of unemployment; production from potential output.
- ▶ Because expectations are hardly ever persistently wrong, production equals output potential in the medium-run.
- ▶ If production is above medium-run potential, wages (and thus prices of goods) rise. → see discussion on slide 5.
 - ▶ We will discuss inflation in the next lecture.
- ▶ Preparation for lecture 6: Slides and chapters 8, 23.2, 23.3 in the textbook

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